

Rising Waters: A Machine Learning Approach to Flood Prediction

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] 1.Disaster Management Authorities 2.Meteorological Departments 3.Local Government Bodies 4.Emergency Response Teams 5.Residents of Flood-Prone Areas 6.NGOs involved in Disaster Relief	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] 1.Limited access to advanced prediction tools. 2.Budget constraints for deploying large-scale systems. 3.Lack of technical expertise. 4.Dependence on internet connectivity. 5.Incomplete historical weather records.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] 1.Traditional Weather Forecasting. 2.Satellite Monitoring Systems. 3.Manual Flood Monitoring. 4.Government Alert Systems. 5.Weather News Channels.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] 1.Unexpected flood occurrences causing loss of life and property. 2.Delayed flood warnings during monsoon seasons. 3.Difficulty in identifying high-risk regions. 4.Insufficient time for evacuation and resource	9. PROBLEM ROOT / CAUSE [RC] 1.Extreme rainfall and climate changes. 2.Lack of accurate early-warning systems. 3.Inadequate flood risk assessment methods.	7. BEHAVIOR 1.Regular monitoring of weather updates during monsoon. 2.High dependence on government warnings. 3.Immediate response during severe rainfall events. 4.Frequent checking of flood risk information in vulnerable areas.

allocation. 5.Frequent flooding in vulnerable regions every year.	4.Delayed communication of disaster alerts. 5.Limited use of machine learning in flood forecasting.	
3. TRIGGERS TO ACT [TR] 1.Heavy rainfall forecasts. 2.Sudden increase in cloud visibility and weather anomalies. 3.Rising water levels in rivers and reservoirs. 4.Monsoon season alerts. 5.Historical flood patterns indicating risk.	10. YOUR SOLUTION [SL] The proposed solution uses Machine Learning algorithms such as Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost to analyze weather and rainfall data. The system predicts flood risk based on factors such as annual rainfall, seasonal rainfall, and cloud visibility. Key Features: • Early flood risk prediction. • Real-time flood classification. • High accuracy prediction using XGBoost (96.55%). • User-friendly Flask web application. • Cloud deployment support through IBM Cloud. • Faster disaster response and evacuation planning. Value Delivered: • Reduces flood-	8. CHANNELS OF BEHAVIOR [CH] ONLINE 1.Weather Websites 2.Mobile Applications 3.Government Portals 4.Social Media Alerts 5.Cloud-Based Monitoring Systems
4. EMOTIONS BEFORE / AFTER [EM] • Fear and uncertainty about floods. • Anxiety regarding public safety. • Lack of confidence in traditional forecasting methods. After Using System • Improved preparedness. • Better confidence in decision-making. • Reduced panic through early warnings. • Increased public safety awareness.		OFFLINE 1.Television News 2.Radio Broadcasts 3.Public Announcement Systems 4.Local Government Notifications 5.Community Awareness Programs

related damages.

- Improves disaster preparedness.
- Supports efficient resource allocation.
- Provides reliable and scalable flood forecasting.